

Adroit OPC UA Client Protocol Driver

Contents

Overview	2
Connectivity	2
Driver Installation and Setup	2
OPC UA Instance Certificates	2
OPC UA Server Certificates	3
Device Configuration	3
Registry details	6
Scanning Configuration	6
Message protocol.....	7
Troubleshooting.....	7
Revisions	8



Overview

OPC UA (OLE for Process Control Unified Architecture) is a standard specification that makes it possible for automation/control software applications and field systems/devices to interoperate and communicate data between each other.

To learn more about what OPC UA is, visit the OPC foundation website
<https://opcfoundation.org/about/opc-technologies/opc-ua/>

The OPC UA client driver for Adroit communicates to OPC servers that expose an OPC UA interface.

Connectivity

The OPC UA client driver communicates with OPC UA Servers in accordance with the OPC UA Specification whose details are beyond the intended scope of this document.

Driver Installation and Setup

If the Adroit OPC UA Client driver is not installed, you will need to install it. Installation of this driver for the first time will install the OPC Foundation's OPC UA redistributables if not already installed on your PC.

The following files will be installed agent server folder:

- d_opc_ua.dll
- d_opc_ua.doc
- opcuabrowser.exe

as well as the following 3rd party dependency files:

- libeay32.dll
- libxml2.dll
- TB5CPP.dll
- TB5OT.dll
- TB5STACK.dll
- TB5UTIL.dll

If any of these files is not in the agent directory, then the driver did not install properly.

OPC UA Instance Certificates

The OPC UA specification requires that all UA clients have an instance certificate that positively identifies client to the server. The instance certificate for Adroit OPC UA is automatically created on the agent server PC when the OPC UA Protocol driver DLL is loaded by PSEXEC or AS.EXE for the first time.

The instance certificate can also be created manually by executing the *%ProgramData%\Adroit Technologies\Adroit\certs\create_instance_certificate.bat* file which is found in the folder. Once the batch file has completed, the following folders will have been created under the *certs* folder:

cert
issuer
own
private

rejected
trusted

The *cert_adroit_client.der* file will be found in the *certs/own* folder and the *private_key_adroit_client.pem* file will be found in the *certs/private* folder. Select these two files for the instance certificate entries above.

Furthermore, the *cert_adroit_client.der* file needs to be copied to the trusted certificate folder of the OPC UA Server/s that you will be communicating to. Conversely, the OPC UA Server instance certificate needs to be copied to the Adroit system folder for two way authentication. The certificate types should be the same.

OPC UA Server Certificates

The OPC UA specification requires that all UA clients trust the server certificate for the OPC UA server that they are connecting to. By default, server certificates will be placed in the *%ProgramData%\Adroit Technologies\Adroit\rejected* folder when a client connects to the OPC UA server for the first time. These must be manually copied to the *%ProgramData%\Adroit Technologies\Adroit\trusted* folder before a successful connection will be allowed.

Device Configuration

When configuring an OPC UA client device, the following dialog is launched and must be completed accordingly.

The dialog has 3 different modes depending on the authentication type selected. By default, the dialog looks as follows for anonymous authentication:

Property	Value
OPC UA Server	
Endpoint URL	
Name	Unknown
Authentication (token) type	<Unknown EnumUserTokenType value>
Message mode	None
Policy	None
General	
Timestamp source	Server
Polled	False
Timeout (ms)	3000
Retries	3
Test connection	
Test	

Username authentication looks as follows:

The screenshot shows the 'OPC UA Client Protocol Driver - UNK' dialog box. At the top, there are radio buttons for 'Primary' (selected) and 'Secondary', and a checkbox for 'Enable secondary'. Below this is the 'Primary connection parameters' section with checkboxes for 'Enable shadow scanning from master to standby', 'Enable standby reads' (checked), and 'Enable standby writes'. A table lists properties and their values:

Property	Value
OPC UA Server	
Endpoint URL	opc.tcp://mikel-pc:48010
Name	UaServerCpp
Authentication (token) type	USERNAME_1
Message mode	None
Policy	http://opcfoundation.org/UA/SecurityPolicy#None
User credentials (if authentication type = Username)	
Username	
Password	
General	
Timestamp source	Server
Polled	False
Timeout (ms)	3000
Retries	3
Test connection	
Test	

At the bottom, there is a red box with the text 'Endpoint URL' and 'Select the OPC UA Server'. 'Cancel' and 'Close' buttons are at the bottom right.

Certificate authentication looks as follows:

The screenshot shows the 'OPC UA Client Protocol Driver - UNK' dialog box with the same layout as the previous one, but with 'CERTIFICATE_2' selected for 'Authentication (token) type'. The 'User certificate (if authentication type = Certificate)' section is expanded, showing fields for 'Certificate file', 'Private key file', and 'Private key password', all of which are empty. The 'General' and 'Test connection' sections are also visible. The red box at the bottom still says 'Endpoint URL' and 'Select the OPC UA Server'. 'Cancel' and 'Close' buttons are at the bottom right.

On selecting the [...] button for 'Endpoint URL' you will be presented with the "Connect to Server" dialog discussed later.

User credentials - if authentication type = *username*, a username and password must be specified

User certificates - if authentication type = *certificate*, a certificate must be specified. User certificates need to be issued by a 3rd party certification authority.

Timestamp source – Local, Server or Source. Default is Server

- Server – When the OPC server sends a value the driver, whether by exception or from being polled, the OPC server return two timestamps, the timestamp that the server received the value, or the timestamp that the

device/source received the value, by selecting this radio button, the driver will update values with the servers timestamp.

- Source – The driver will update values with the device/source timestamp as mentioned above. (NB, do not use this timestamp option if the communication protocol connecting to the device does not supply a timestamp)
- Local – If this option is selected, the driver will update values with the timestamp of when the driver receives the value.

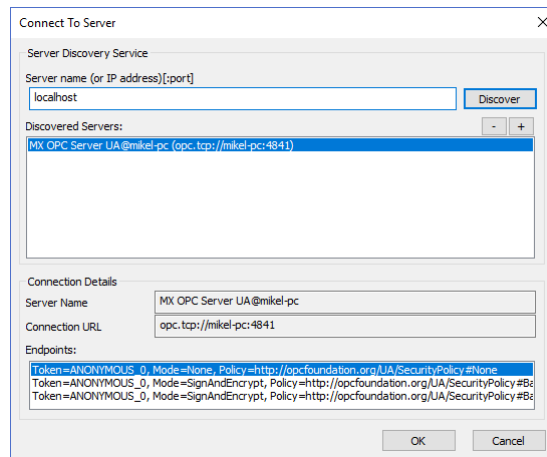
Polled – OPC clients by nature receive data “by exception” from OPC Servers. However, you can use OPC for polling data from front end devices. This setting enables polling as opposed to “by exception”. This setting should only be used if absolutely necessary. (Default off)

Timeout – Poll and connection timeouts in msec

Retries – Number of retries before failing a poll transaction

Test – Test the connection to the OPC UA server using the configured endpoints. If the connection to the OPC UA server is being made for the first time, you will be prompted as to whether you wish to trust the server certificate.

Connect To Server



- Server name / Server IP address – This is the name or IP address of the machine on which an OPC UA server discovery will be conducted. To initiate a discovery on the machine or IP address entered, click on the refresh button.
- Discovered Servers – This list displays all available OPC UA servers which are discoverable on the machine or IP address specified in “Discovery Host Machine”.

The [+] button allows you to enter a server manually that is not discoverable on the network if the format *opc.tcp://hostname:port* e.g. *opc.tcp://klaus:1234*. Manual entries are stored as a CSV entry in the registry value HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\OPC UA\Manual.

The [-] button allows you to remove the currently selected server, if it is a manually added server.

If any of the servers displayed in “Discovered OPC Servers” are clicked on, then the “Connection Details” will automatically populate with the selected server’s details.

Connections Details

- Server Name – The name of the server to which the current device will connect to.
- Connection URL – The server endpoint address on which the device will connect when connecting to the OPC UA server.
- Endpoints – The list of endpoints (authentication methods) supported by the selected OPC UA Server
- Message security mode
- Security policy

The combination of token type, message security mode and security policy must be valid for the OK button to be enabled.

NOTE – It is best to always make your selection and press [Ok] on this “Connect to Server” dialog because pressing [Cancel] will make all the “OPC UA Server” group of properties editable. Editing these properties manually is done at your own risk.

Registry details

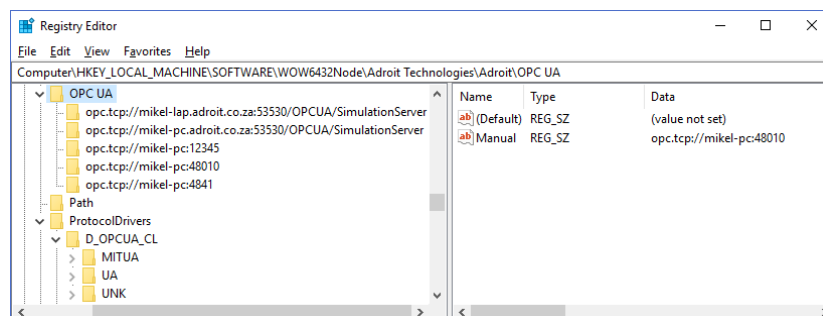
As is the case with all Adroit Protocol Drivers the configuration for a device is stored in the registry at

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\D_OPCUA_CL\DeviceName

However, to allow both the driver and opcuabrowser.exe to share connection details for OPC UA Servers, these details are stored at

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\OPC UA\ConnectionURL

An example of the registry store is show below where both registry sections are visible:



Scanning Configuration

The default address is $ns=1;s=address$.

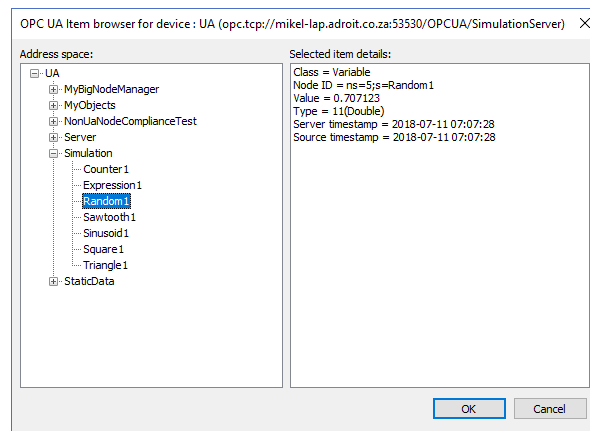
Each node in the OPC UA address tree is made up of an address tuple which contains the namespace index and address of the node.

The OPC UA address format is as follows:

$ns=x;s=y$

Where x is the namespace index of the node and y is the address of the node respectively.

A node's namespace index and address can found/browsed by using the OPC UA Browser. Clicking the browse button in the scanning dialog will launch an item browser (not opcuabrowser.exe!) which will browse the entire address tree on the connected OPC UA server. The browser dialog looks as follows:



If the driver is in polled mode, the update rate of a tag is determined by the scan period specified.

Please note :

1. This driver does not support dula scanning. In other words should you scan more than one tag to a single address, subsequent tags will always remain scan bad.
2. This driver does not support "Output Initialize".

Message protocol

Datascope messages are displayed in ASCII mode.

The underlying communication protocols to the device is not available to the OPC Client and thus not showed, see the [troubleshooting](#) section.

This window will however show all validation of tags. If the tag cannot be validated, a message indicating the error message will be displayed here.

All item updates (process-data changes) are also indicated here with an offset number of the tag that's being updated.

Troubleshooting

Datascope errors

Cannot connect to server	Ensure that the OPC UA Server is running
Re-connect failed (ERROR : Failed to get endpoints: 0x80AD0000(BadDisconnect))	Ensure that the OPC UA Server is running
Re-connect failed (ERROR : Failed to connect the session: 0x801A0000(BadCertificateUntrusted))	Ensure that the certificate of the OPC UA Server is trusted by Adroit by moving the certificate from the rejected folder to the trusted folder

Re-connect failed (ERROR : Failed to connect the session: 0x80130000(BadSecurityChecksFailed))	Ensure that the Adroit certificate is trusted by OPC UA Server
Re-connect failed (ERROR : Failed to get endpoints: 0x80AD0000(BadDisconnect))	1. Ensure that the OPC UA Server is running 2. Ensure that the certificate of the remote OPC UA Server is trusted

Message boxes

ERROR : Failed to find servers at opc.tcp://hostname:4840	1. The host does not exist OR 2. The host is not locatable by your DNS Server - try using the IP Address instead of the <i>hostname</i> 3. The OPC UA Local Discovery Service is not running on the host 4. If <i>hostname</i> is the local PC - there is a known issue caused by Windows' preference of IPV6 over IPV4. See https://theitbros.com/ping-returns-ipv6-address-ping-ipv4/ for more information on the subject which will tell you how to set IPV4 over IPV6. Otherwise, simply use the IP address of the host instead of the host name.
The OPC UA Server certificate is not trusted by this server. Would you like to trust this OPC UA Server?	If you press [Yes] Adroit will trust the OPC UA Server for you.
Our client certificate is not trusted by the OPC UA Server. Please invoke the trust on the OPC UA Server (or move the rejected client certificate to the trusted folder on the OPC UA Server)	If you press [Yes] Adroit will trust the OPC UA Server for you.
ERROR: Failed to get endpoints: 0x803D0000(BadNotSupported)	

Revisions

27-May-2015	First Draft
03 July 2018	Added OPC UA browser section
16 October 2018	Removed OPC UA browser section